
Privacy Analysis of Augmented Reality Glasses

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Weekly Report 8

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What has been done (Previous Week)

- Asked during the PET meeting for a sim card that has mobile data.
- Finish a trial/experiment for all Bluetooth traffic captures and uploaded onto gitlab. The following depicts the experiments structure:
 - Classic Bluetooth -
 - * With internet connection - includes initial pairing
 - For touch commands
 - For voice commands
 - * Without internet connection - does not include initial pairing
 - For touch commands
 - For voice commands
 - BTLE -
 - * With internet connection - includes initial pairing
 - For touch commands
 - For voice commands
 - * Without internet connection - does not include initial pairing
 - For touch commands
 - For voice commands
- Successfully set up for a dynamic analysis for the APK of the Meta View app using MobSF though there was a lack of interaction due to the glasses not paired to the emulator on the laptop.
- Written documentation on the set up of the static and dynamic analysis (using Genymotion for the emulation) that still needs to be improved and uploaded.
- Started with rooting the phone, however, encountered a problem with unlocking the Mi Account due to the sim card. The step that needs the sim card is for unlocking the bootloader.
- Went through the description of the thesis and have a small documentation on what can be done with the current results I have as of now and what I would need to finish the rest.
- Set up mitmproxy on my laptop

What will be done next week (and following weeks as some tasks may extend over multiple weeks)

- Continue to root the phone (once received the sim card) and if so write documentation about the process.
- Write documentation on Bluetooth traffic experiments done the previous weeks.
- Finish the documentation for the set up for the dynamic APK (also try to see about the pairing for the glasses and the emulator).
- Use the static/dynamic analysis and write documentation in regards to some of the goals for the analysis (e.g. third party libraries, permissions, etc.) or anything else interesting for us.
- Try to use mitmproxy for analyzing network traffic (e.g. experimenting with TLS inspection to analyze the traffic or identify some patterns through metadata).
- Try to decrypt traffic without rooting the phone (but still continue trying to root the phone on the side as said above).
- Document any result or analysis that can be done on the encrypted traffic from the experiments done in regards to the thesis goals (e.g. identifying third party destinations, or traffic patterns, etc).
- Since I have the new phone, I will try to see if I can get the HCI logs from this phone for the Bluetooth traffic.

Meeting

Meeting canceled.